

ASSIGNMENT: OPERATING SYSTEMS

Course Code: INSY 8223

Total Marks: 30

Submission: Individual

Instructions: Answer all questions clearly. Use examples where necessary.

SECTION A: Lesson 1 – Introduction to Operating Systems (10 Marks)

Question 1 (5 marks)

- a) Define an **Operating System**.
- b) Explain **any four (4) major functions** of an Operating System.

Question 2 (5 marks)

Using clear explanations, differentiate between the following types of Operating Systems:

- a. Batch Operating System
- b. Time-Sharing Operating System
- c. Real-Time Operating System

SECTION B: System Calls and OS Security (12 Marks)

Question 3 (6 marks)

- a) What are **System Calls**?
- b) Explain **three (3) categories of system calls**, giving **one example** for each category.

Question 4 (6 marks)

Explain how these services improve system performance and usability.

Describe the **Operating System services** provided to:

- a) Users
- b) Programs

SECTION C: Critical Thinking & Application (8 Marks)

Question 5 (4 marks)

Explain the role of an Operating System in **system security**.
Discuss **any two (2) security mechanisms** used by operating systems.

Question 6 (4 marks)

Assume you are designing an Operating System for a **banking system**.

- a) Which OS features would be most important?
- b) Justify your answer.